Soutenance
Projet Drone

deuxième semestre
Groupe III

responsables par domaine

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1- Cahier des charges fonctionnel (drone) (MEO)
2- Cahier des charges technique (MEC) (bras du drone)
3- Calibration et implémentation (caméra) (EEE)
4- Principe de vol et choix matériaux (EE
5- Follow me (MIA)
6- Impact du drone sur l'humanité (ESE)

01

Outil de **gestion de projet**

(MEO) Cahier de charge

❑ Présentation du projet

Contexte

L'industrie des drones connaît une croissance rapide et transforme divers secteurs, allant de la surveillance environnementale à la livraison de colis, en passant par l'agriculture de précision et la photographie aérienne. La maîtrise de cette technologie représente un atout considérable pour les futurs ingénieurs, qui doivent être capables de comprendre et de manipuler des systèmes complexes combinant mécanique, électronique, informatique embarquée et dynamique de vol.

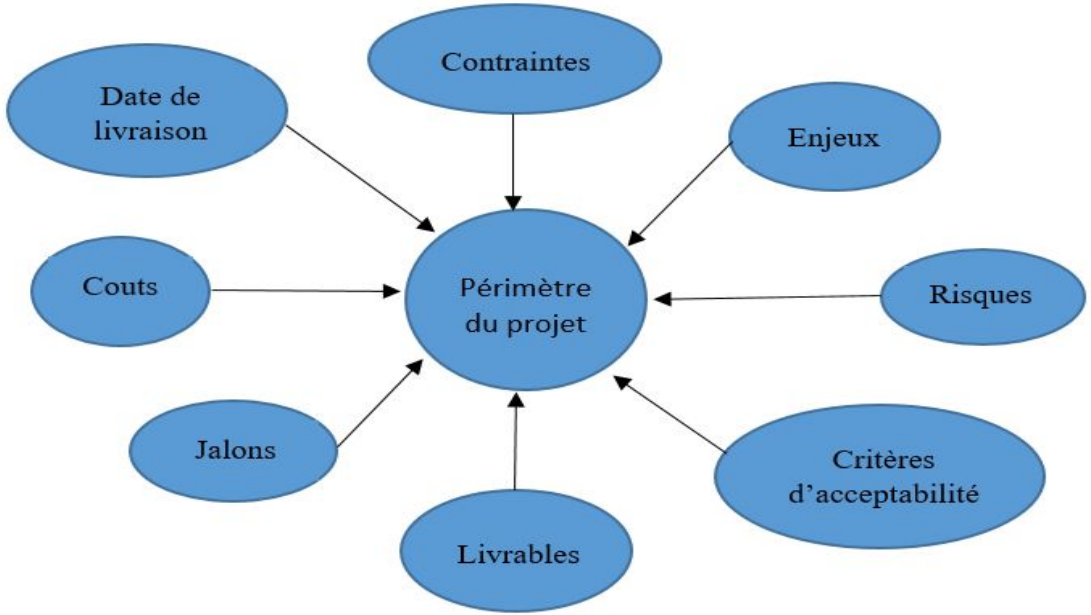
Objectif du Projet

L'objectif principal de ce projet est double. D'une part, il vise à développer nos compétences techniques et pratiques à travers le montage, la configuration et le pilotage d'un drone. D'autre part, il cherche à nous familiariser avec la gestion de projets complexes en ingénierie, en intégrant des éléments de planification, de travail en équipe et de résolution de problèmes.

❑ Organisation et Responsabilité

Périmètre

Le périmètre du projet définit les limites et les étendues des activités à réaliser pour atteindre les objectifs fixés.



❑ BESOINS ET CONTRAINTES

Critères d’acceptabilité

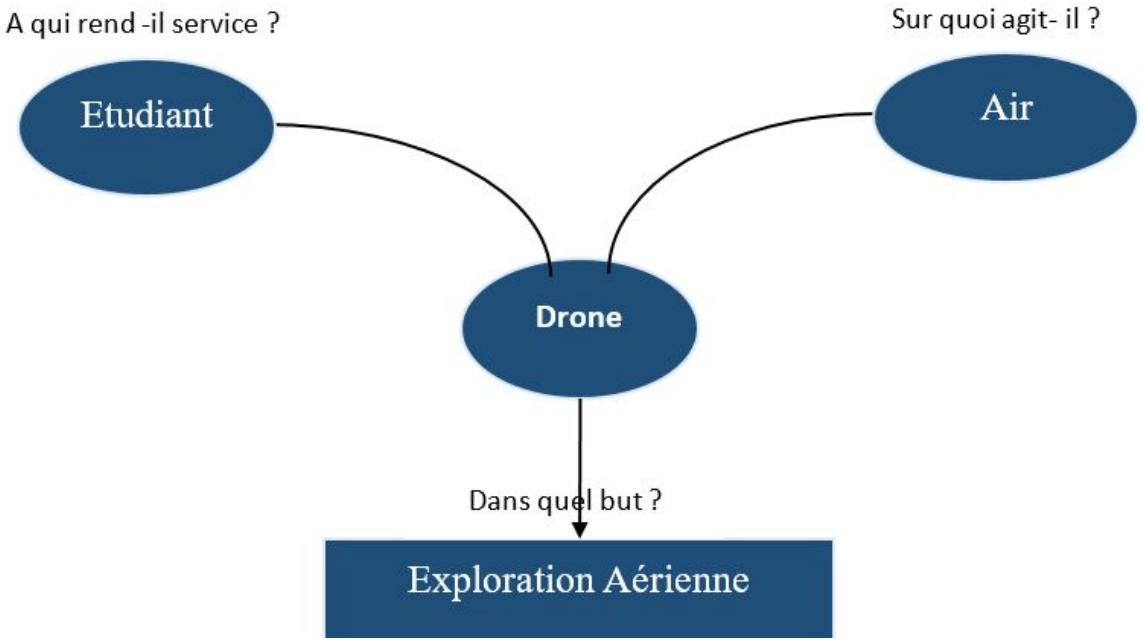
1. Bonne organisation: Une structure bien définie des tâches et des responsabilités.
2. Respect du temp et du déroule succinct: Bonne gestion du temps et Suivi rigoureux du planning de travail.
3. Qualité de la main-d'œuvre: Compétences adéquates et engagement des membres du groupe.

Risques

- Pannes de matérielles ou dysfonctionnements.
- Mauvaise communication interne et organisation.
- Retards sur le dépôt des livrables.
- Non-conformité aux normes de bonne conduite et sécurité.
- Defaillance de la main d’oeuvre.

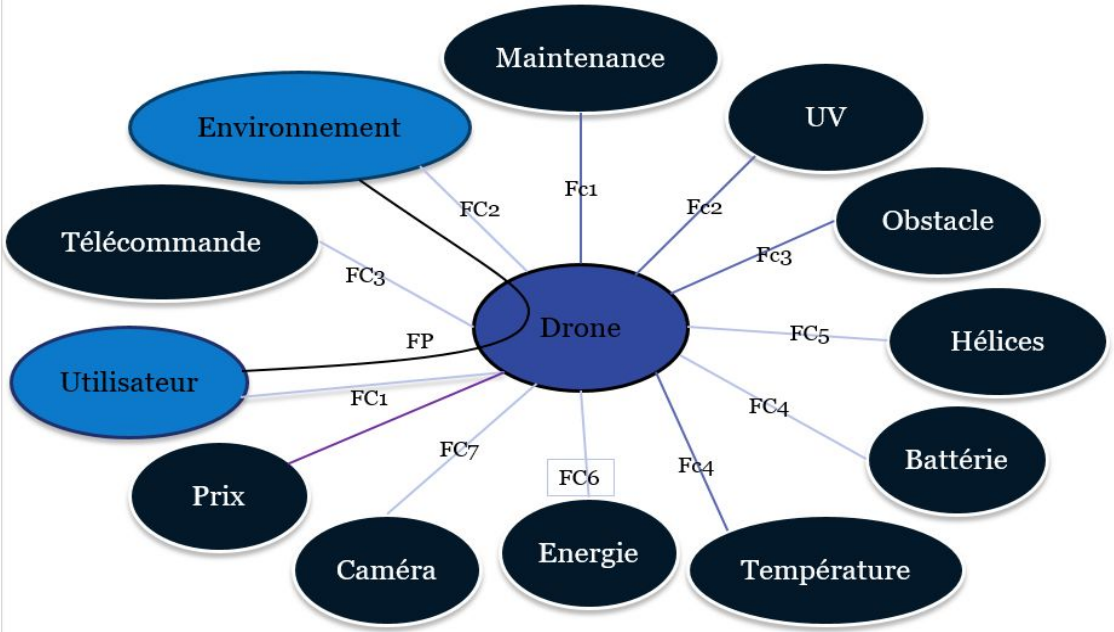
Besoin Fonctionnel (Bête à Corne **)**

Ici, la bête à corne met en évidence notre besoin fondamental en répondant à certaines questions.

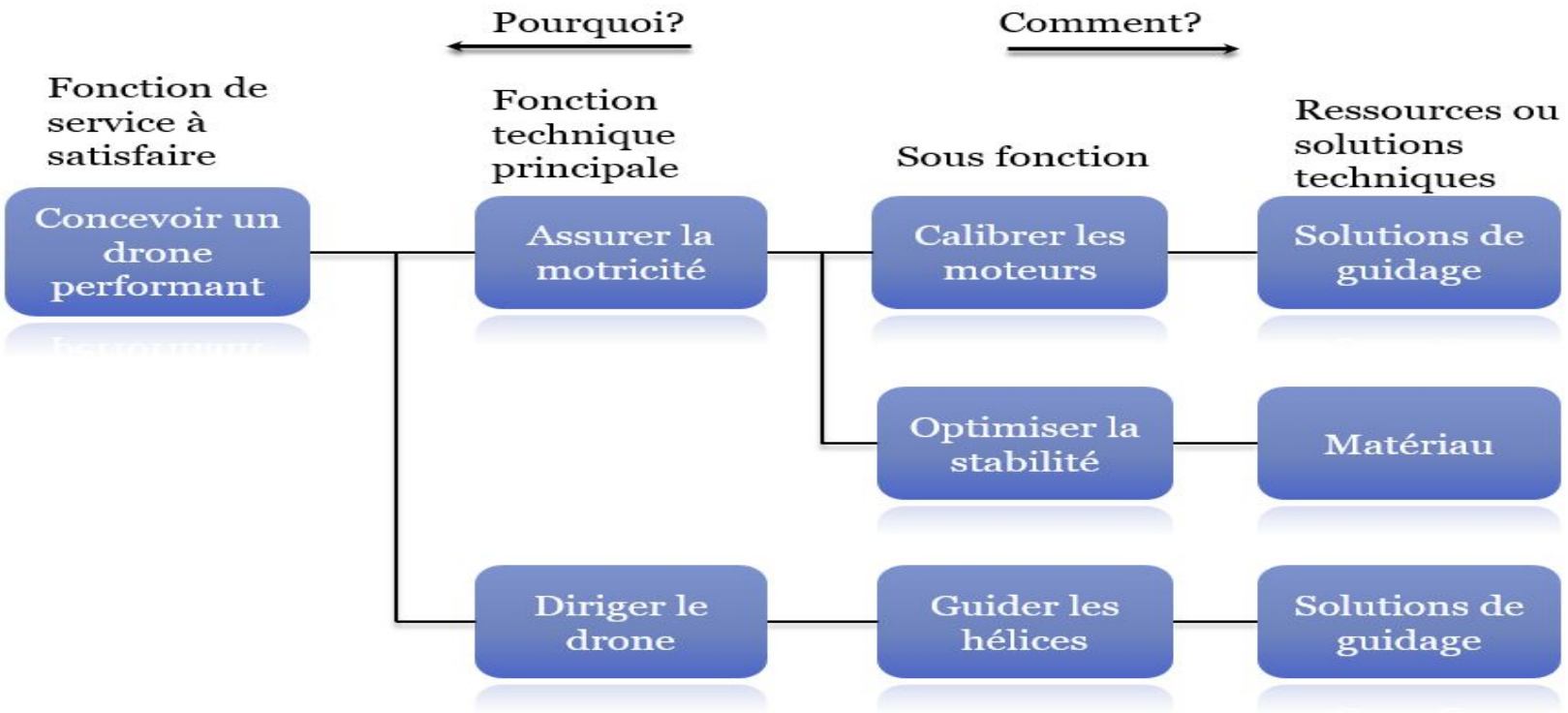


❑ CAHIER DE CHARGES FONCTIONNEL

Description des fonctions



➢ Méthode Fast

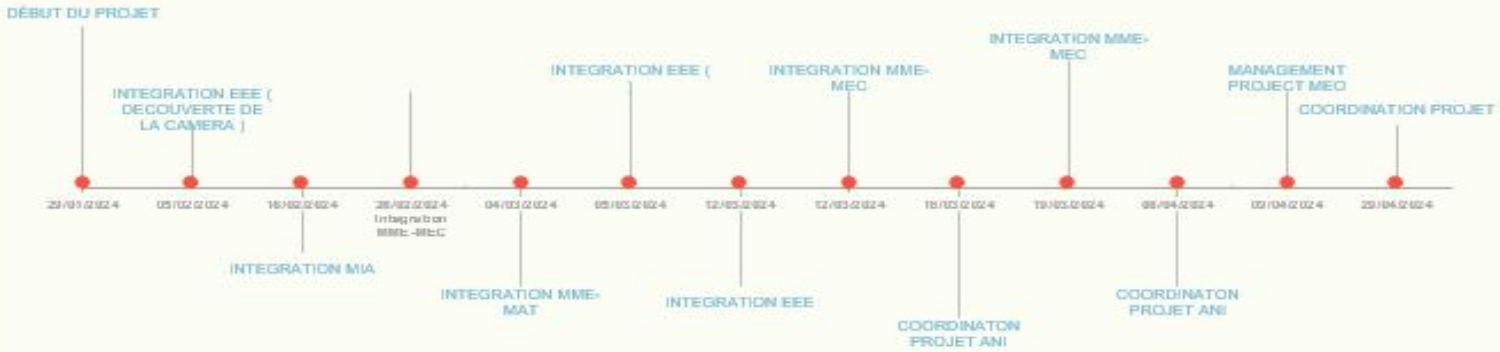


❑ COÛTS

Pièce	Prix (€)	Référence d'achat
Châssis principal F450	15	myhelis.com
Bras rouge de rechange	8	hobbywireless.com
Bras blanc de rechange	8	myhelis.com
Contrôleur de vol Naza-M V2	229	rcsuperstore.com
Moteur 2212 920 KV	19	myhelis.com
ESC 30A OPTO	14	myhelis.com
Hélices 9450 (paire)	8	hobbywireless.com
Kit ARF (sans contrôleur)	190	rcsuperstore.com

VOTRE PROJET

CHRONOLOGIE



DÉTAILS DU PROJET

DATE	JALON	POSITION
29/01/2024	Début du projet	25
05/02/2024	Integration EEE (Decouverte de la car	10
16/02/2024	Integration MIA	-10
26/02/2024	Integration MME-MEC	15
04/03/2024	Integration MME-MAT	-15
05/03/2024	Integration EEE ( )	15
12/03/2024	Integration EEE	-15
12/03/2024	Integration MME-MEC	15
18/03/2024	Coordinaton projet ANI	-20
19/03/2024	Integration MME-MEC	20
08/04/2024	Coordinaton projet ANI	-15
09/04/2024	Management project MEO	15
29/04/2024	Coordination Projet	10
30/04/2024	Integration MIA	-10
18/05/2024	Integration EEE	15
29/05/2024	Fin (Soutenance projet)	-15

Astuces de chronologie de projet :

Le rôle des valeurs Position dans le tableau Détails du projet est d'empêcher les indicateurs de jalons de se chevaucher sur la chronologie. Utilisez des nombres positifs pour placer les indicateurs au-dessus de la chronologie et des nombres négatifs pour les placer en dessous.

Pour ajouter des jalons supplémentaires, insérez de nouvelles lignes dans le tableau ou commencez à taper en dessous de la dernière entrée du tableau. Celui-ci s'agrandira automatiquement pour recevoir les nouvelles données ajoutées.

02

cahier des charges techniques et optimisation du bras

domaine MEC

INTRODUCTION

Le projet de conception des bras de drone DJI F450 est un projet technique qui nécessite une planification minutieuse et une compréhension claire des objectifs visant à développer des bras de drone légers et efficaces pour le modèle DJI F450. Ce projet s'inscrit dans le cadre plus large de l'innovation en matière de drones, avec l'objectif d'améliorer la performance et la fiabilité des drones DJI F450.



Bête à corne

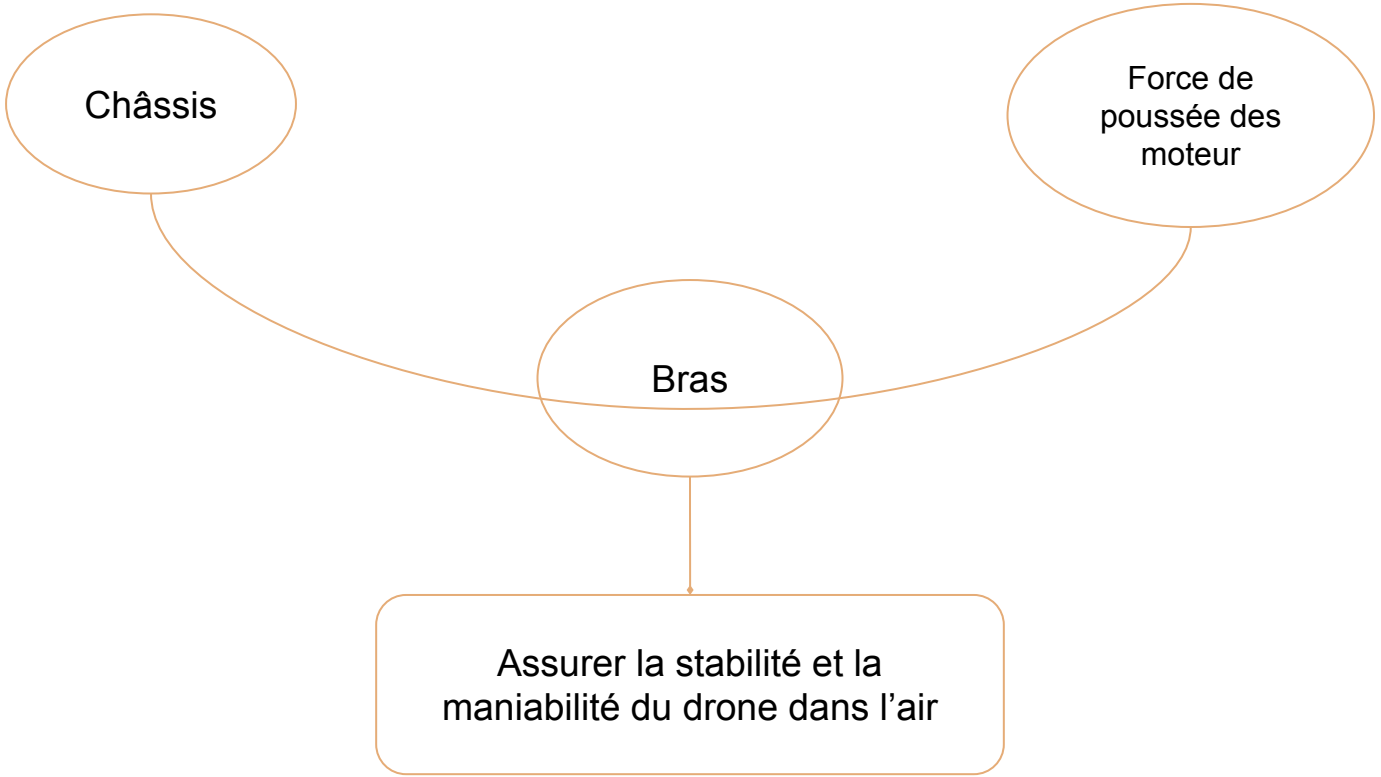

Châssis
Force de poussée des moteur
Bras
Assurer la stabilité et la maniabilité du drone dans l'air

Diagramme pieuvre

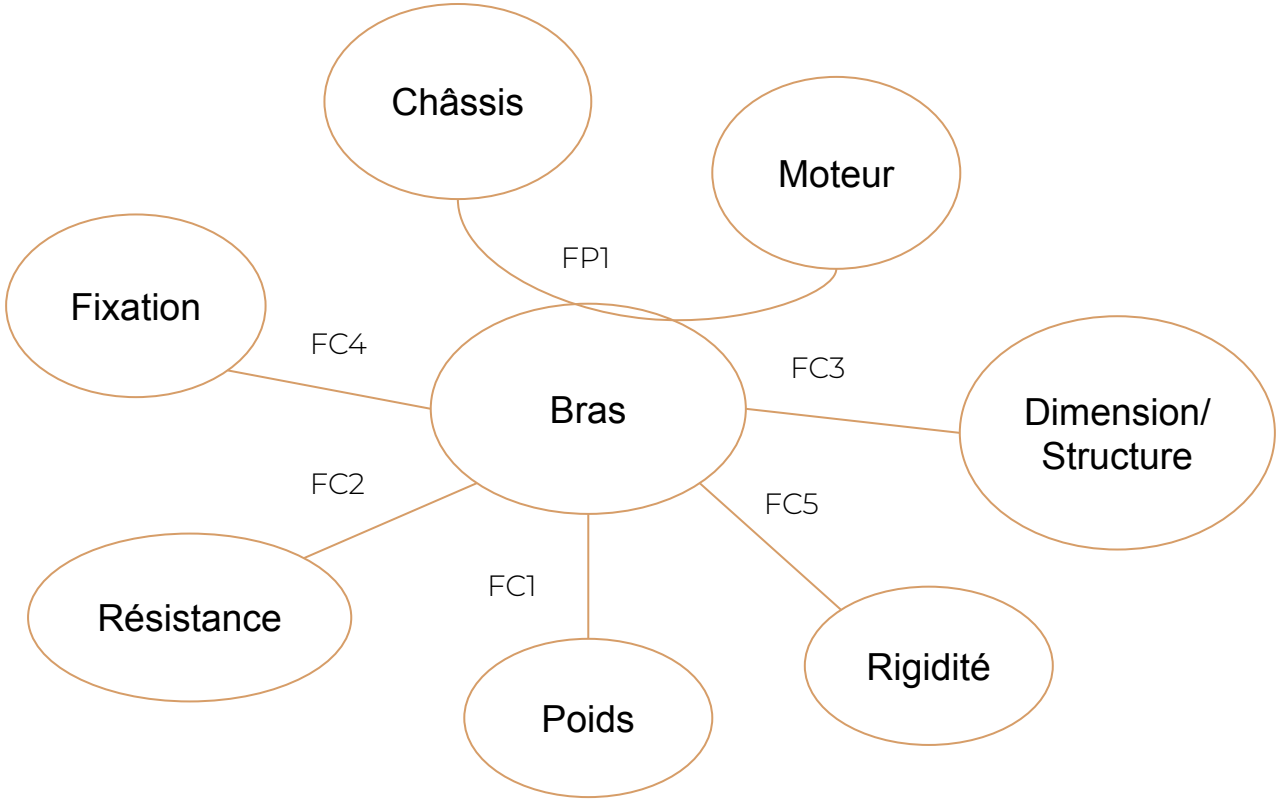
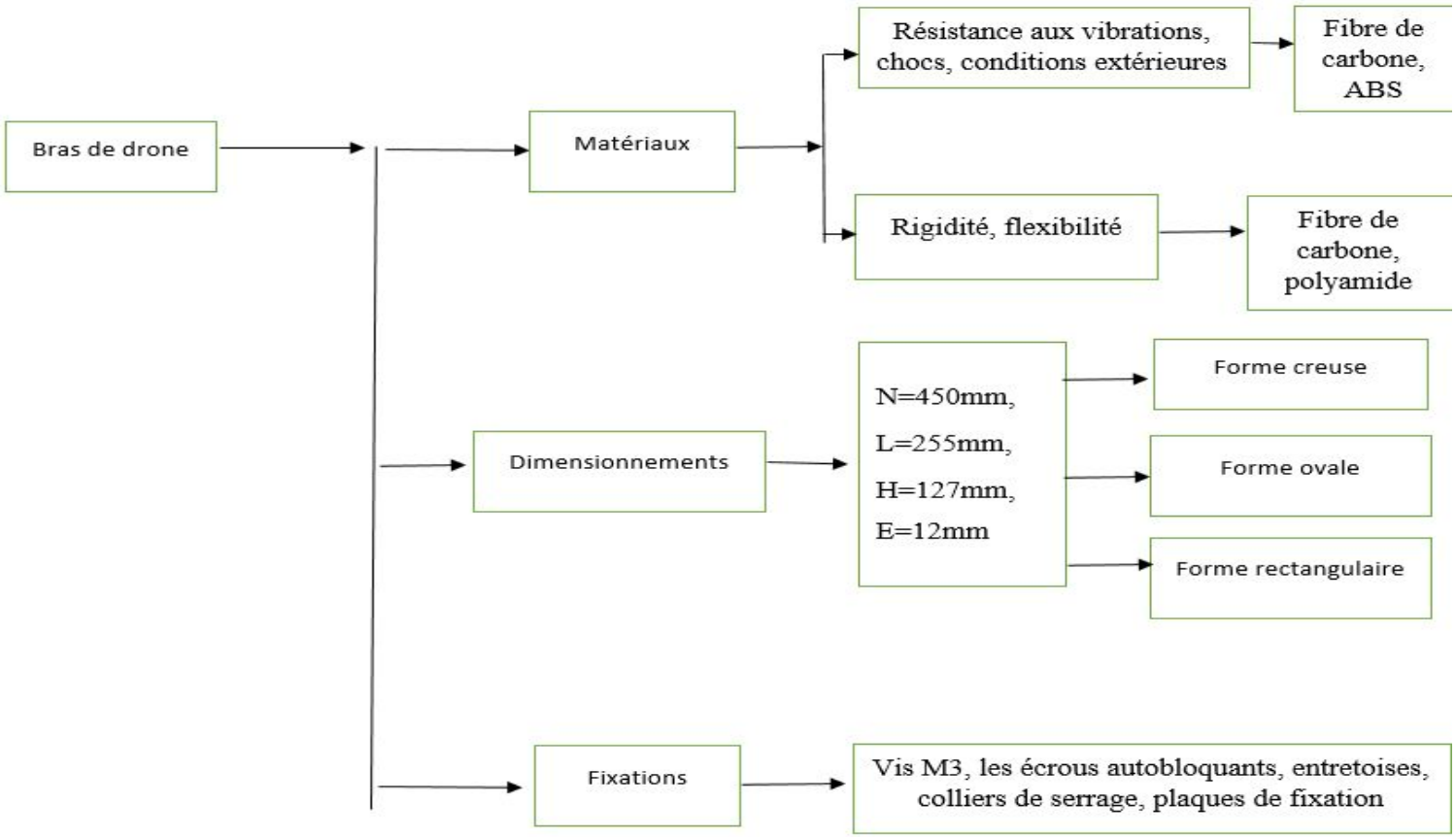


Tableau fonctionnel

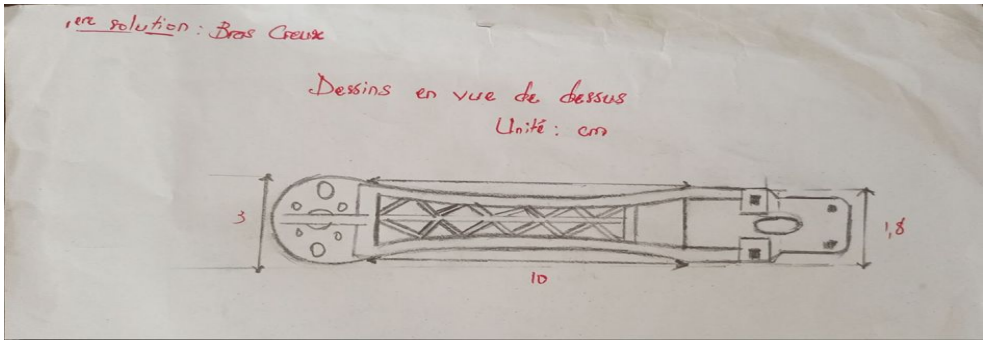
Fonction	Description	Critères	Niveau	Flexibilité	Justification
FP1	Excentré les moteurs pour une rotation libre des hélices	Positionnement précis des moteurs	X-Frame, H-Frame, O-Frame, V-Frame	F1	Permet un vol équilibré et réactif
FC1	Optimiser la stabilité tout en minimisant la masse	Stabilité, poids léger	282 gramme 50 g et 60 g	F1	Performances de vol améliorées
FC2	Résister aux contraintes mécaniques et environnementales	Résistance aux vibrations, chocs, conditions extérieures	Matériaux Robuste et durable (fibre de carbone, ABS)	F2	Fiabilité et durabilité accrues
FC3	Définir les mesures pour faciliter la maniabilité du drone	Dimensions appropriées pour la manœuvrabilité	N=450mm, L=255mm, H=127mm, E=12mm	F3	Réactivité et manœuvrabilité améliorées
FC4	Fixation solide, simple et démontable des bras au châssis et aux moteurs	Fiabilité de la fixation, facilité de démontage/remontage	Vis M3, les écrous autobloquants, entretoises, colliers de serrage, plaques de fixation	F2	Réduction du temps d'entretien et des coûts
FC5	Maintenir la forme sous contrainte tout en étant suffisamment flexible pour permettre des mouvements	Rigidité, flexibilité	Équilibre entre rigidité et flexibilité (fibre de carbone, polyamide)	F2	Réduction des risques de déformation et d'endommagement

Diagramme fast

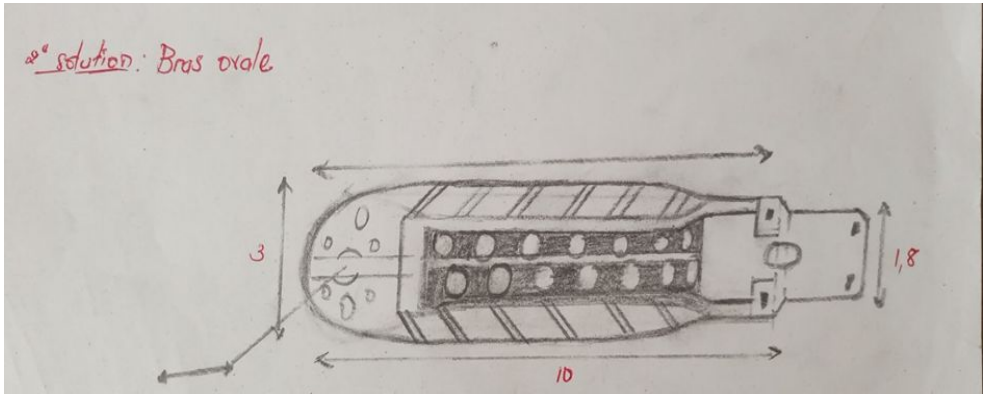


2-Rapport sur l'Optimisation du Bras

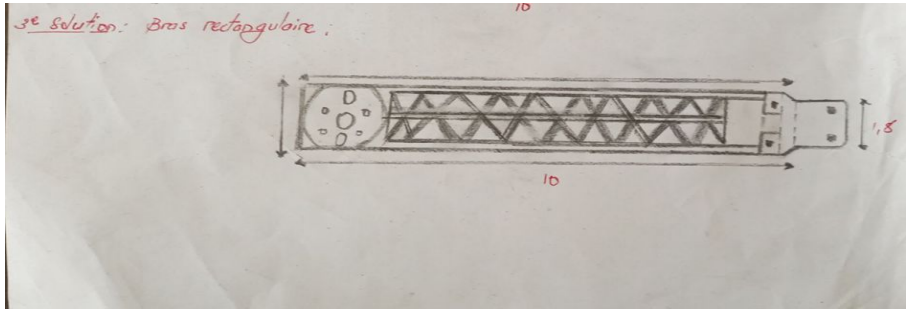
Bras creux :



Bras ovale :



Bras rectangulaire :



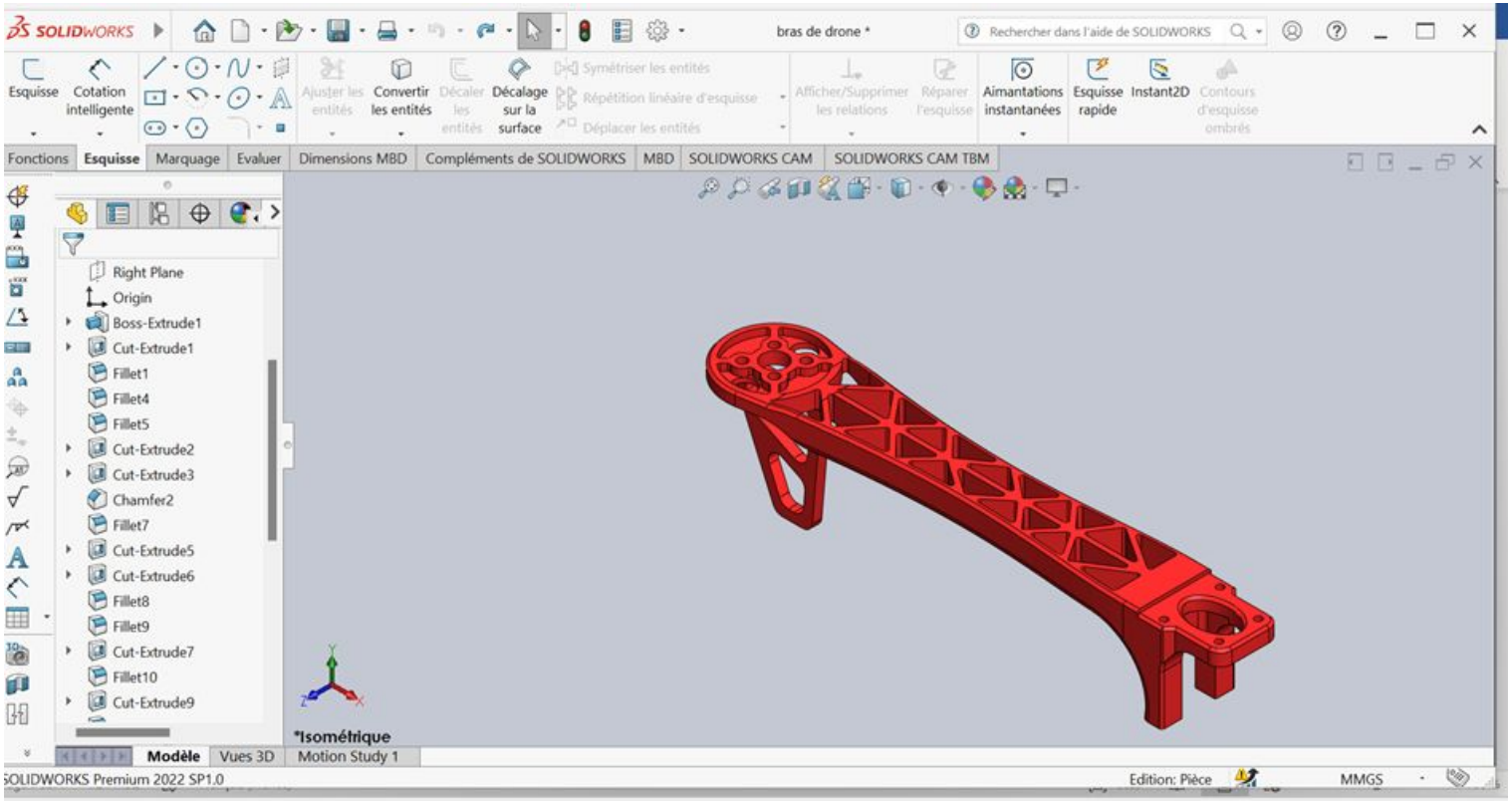
Critique Constructive :

Chaque conception a ses avantages et ses inconvénients. Le bras ovale est résistant mais peut être plus lourd. Le bras creux est léger mais peut-être moins résistant aux forces de flexion. Le bras rectangulaire est facile à fabriquer mais peut-être moins résistant aux forces de torsion. Une analyse plus approfondie est nécessaire pour déterminer quelle conception offre le meilleur équilibre entre la réduction de la masse et la conservation de la résistance mécanique.

MATRICE DE DÉCISION

Critèsres	Coefficient	Bras Ovale	Bras Creux	Bras Rectangulaire
Poids	10	7	9	8
Résistance mécanique	8	8	7	7
Cout	7	6	8	7
Rigidité	6	8	7	7
Dimensions structurelles	4	7	8	8
Fixation	2	8	7	8
Score Total	-	268	290	275

Dessin du bras de drone sur solidworks :



03

Implémentation et calibration

DOMAINE : EEE

-Implementation of a camera
-Establish all the different sensors in the drone
-calculation involving the battery and its Autonomy

Domaine EEE

INTRODUCTION

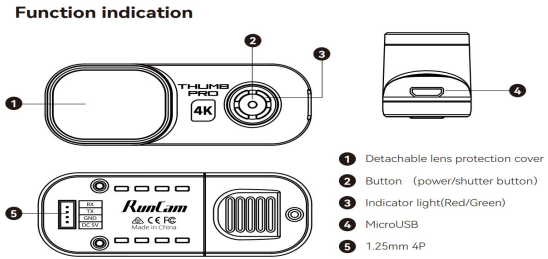
The EEE domain allows us to analyze the electronic system of the drone point by point in order to better understand how it works. But still going to review them back in the final power point respectively.

But for this 2nd semester, the work was carried out as part of the drone project in the field of electrical engineering (EEE) concerning the objectives of semester 2. The main deliverables include the addition of a shooting camera on the drone, the transmission and means of viewing the images, the procedure for calibrating sensors and ESCs, as well as the refinement of the calculation of battery life.

This report covers phases 1 and 2 of the Electrical Engineering field, detailing the steps of implementing the camera, the calibration of the sensors and the calculation of the autonomy of the drone in flight

PHASE 1: Characteristics and implementation of the camera on the drone

Before the characteristics we are first of all going to talk of the description of the camera;







1-Characteristics and specification of the camera

- Resolution: 4K@30fps, 2.7K@60fps, 1440P@60fps, 1080P@120fps, 1080P@60fps
- **Sensor**: IMX 577 (12MP)
- **F.O.V**: 155°
- **Video Format**: MP4
- **Gyro**: Built-In
- Support Micro SD Card capacity:256G Max
- USB Power Input voltage: DC 5V
- **Remote control**: UART
- **Power Consumption**: 500mA at 5V Max
- **Communication Interface**: Micro USB
- **Dimensions**: 54mm x 25.5mm x 21mm
- **Video capture resolution** : 4K
- **Net Weight**: 16 grams



2-IMPLEMENTATION OF THE CAMERA ON THE DRONE

In this phase, our group focused on adding a camera to the drone and its integration.

The steps taken included the study of the fixation of the camera to the drone, the verification of the possibility of shooting and the quality of the images, as well as the test of image transmission.

FIXATION OF THE CAMERA ON THE DRONE

We were expected to be able to move our drone with a photographic camera, which allows the drone to take photos and videos of very good resolution during its flight. To achieve this, we first found the ideal camera and know how to attach/equipped it to the drone. To get there, we chose to use a GIMBAL (CAMERA SUPPORT or NACELLE (in French)).

- **What is a GIMBAL(NACELLE)?**

  The nacelle, also called a drone stabilizer (camera support) is a mechanical device mounted on the drone that allows the camera to be stabilized and controlled

- **How does the GIMBAL works?**

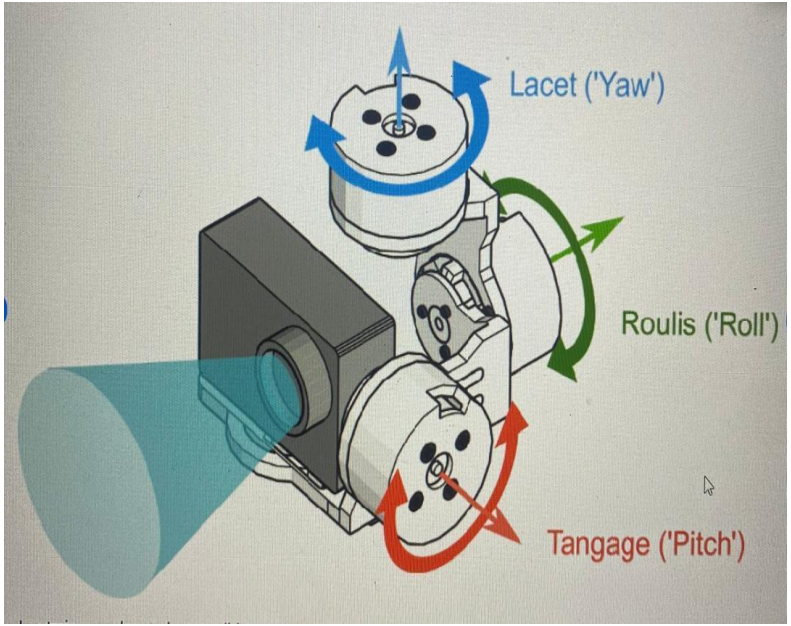
  The nacelle usually uses motors and sensors to compensate for undesirable movements such as vibrations, tremors, inclinations of the drone. The engines are reacting to the signals sent by the sensors and adjusts the position or inclination of the camera to keep it stable.

TYPES OF GIMBALS(NACELLE)

We can find a wide variety of nacelles, but here we are going to talk about the following nacelle;

- **Nacelle with two axes**: it's is a nacelle designed to provide stability to the camera along two axes, making it possible to maintain 1 straight horizon and attenuate unwanted movements along these axes.

- **Nacelle with three axes**: it's is a nacelle designed to provide stability to the camera according to three axes, attenuating unwanted movements along these axes while maintaining a straight horizon





- **Motorized nacelle:** these are nacelles equipped with a motor that allows a controlled movement of the camera. They can be controlled remotely to adjust the angle of the camera during flight, offering great flexibility for image and video recording.



TABLE OF DECISION OF CRITERIA

The following table is a comparative table between different nacelles above to choose the most suitable for our drone.

	Motorized nacelle	**Nacelle with 2-axis**	**Nacelle with 3-axis**
Consumption(v)	10 – 25	7.5 – 11.2	12 – 30
Weight(g)	175 – 500	160 – 500	210 – 960
Stabilization	Good	Good	High
Flexibility	Have compatible software	Good flexibility	Very accessible flexibility
Control	Manual	Automatic/Manual	Automatic/Manual
Cost	115 -400$	70-100$	95-500$

CHOICE OF THE NACELLE

In conclusion, we've chosen the 2-axis nacelle.

REASON FOR THE CHOICE

We chose the 2-axis nacelle due to the fact that it offers good stability and flexibility during flight for quality aerial shots, it has an average weight compatible with small and medium-sized drones and an affordable cost. They can be used for a variety of applications, whether it is professional photography, videography or for inspection (surveillance)

HOW IS THE CAMERA MOUNTED ON THE NACELLE

Before the camera is mounted on the nacelle (Gimbal), its first all place in a **flexible camera support**

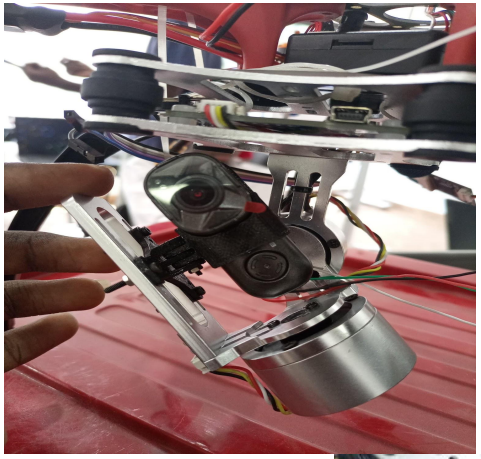




Then now using the flexible camera support, we're going to fix the camera to the nacelle in an equilibrium position and verify the function of the camera. Here it follows;

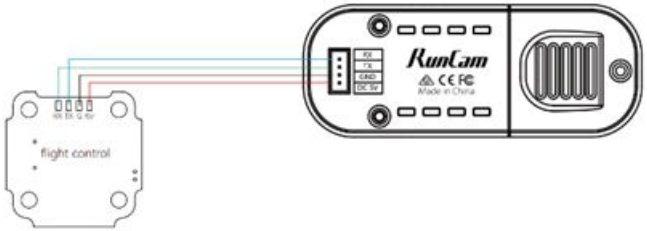






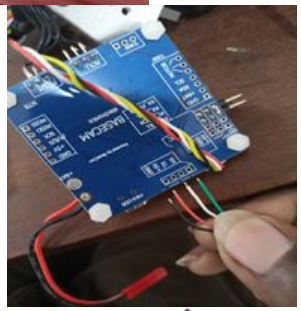
Now, **connecting the camera to the nacelle (camera support);**

Power supply connection and flight controller connection



Please avoid long video recording in the case of high temperature and no ventilation.
Overheating will affect the performance of the device and cause the product to operate abnormally.

Requires welding





PLACING THE NACELLE(GIMBAL) ON THE DRONE ALREADY WITH THE CAMERA

We first have to verify if the nacelle is compatible with the drone, then using attachment we can attached it firmly or screw firmly on the drone. As shown below;



NOW WE MOVE

TO PHASE 2.

PHASE 2: CALIBRATION AND AUTONOMY CALCULATIONS

In this phase, our main objective was to finalize the calibration procedure of the various sensors and ESCs. We carried out extensive tests to ensure the accuracy and reliability of the sensors used. In addition, we used mechanical and energy data to refine the calculation of battery life.

I-CALIBRATION OF SENSORS AND ESCs

To allow the drone to fly correctly and according to the desired directions, we had calibrated some sensors on the drone (sensors for the direction and location of the drone in flight). These sensors are the Compass (boussole) and the accelerometer, we have also calibrated the radio control and the ESCs. To be able to do all these calibrations we used the Mission planner application.

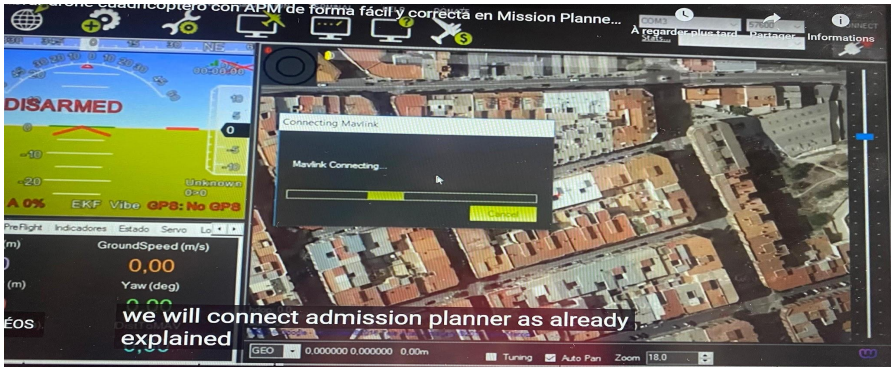
A-COMPASS CALIBRATION

Calibration of the compass for a drone consists of providing accurate information on the magnetic orientation of the drone in relation to the Earth's magnetic field. This allows the drone to know its direction and angle of orientation when flying.

NOTE: DO NOT calibrate the compass in locations where magnetic interference may occur, such as close to magnetite deposits or large metallic structures. Do NOT carry objects (such as cell phones) that contain ferromagnetic materials near the aircraft during calibration.

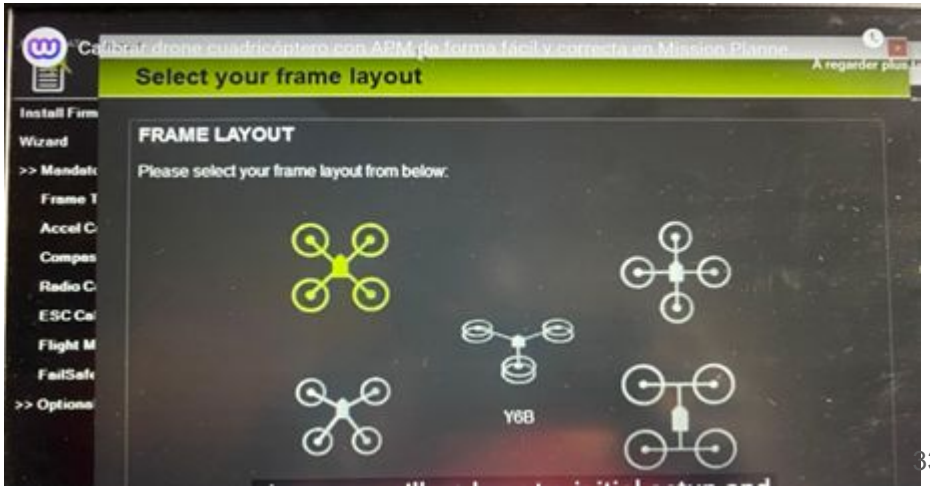
These are the steps to follow for the calibration of the compass, Choose an open area to carry out the following procedure:

1-First connect the Quadricopter to the computer and connect the battery. We will connect the admission planner and start to calibrate.



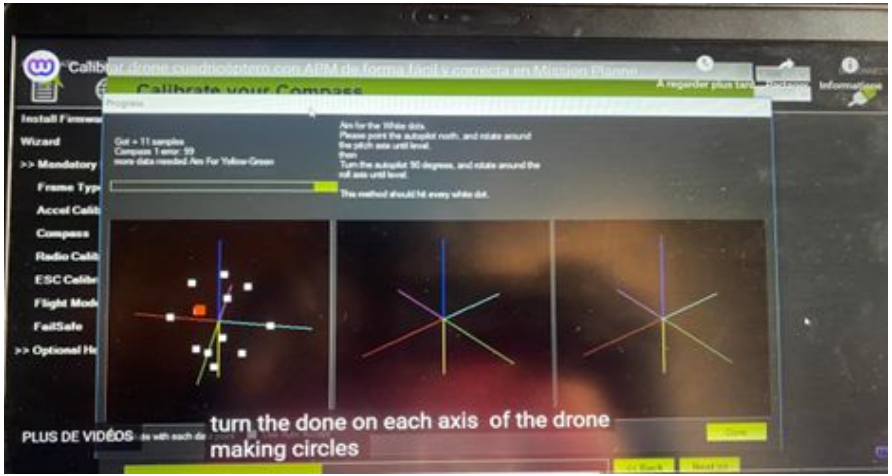
2-Then we move to initial setup and then select **WIZARD**, on the screen will appear different frame and we are going to select the frame in the shape X as a Quadricopter as below;





3-Next we going to select compass calibration and select live calibration then OK, then we have to follow the instructions required. What we have to do is to turn the drone on each axis of the drone making circles in the best possible way that you can.





4-Hold the aircraft horizontally and rotate it 360- degrees. The Aircraft Status Indicator will turn solid green.

5. Hold the aircraft vertically, with its nose pointing downward and after pointing upward, and rotate it 360 degrees around a vertical axis.

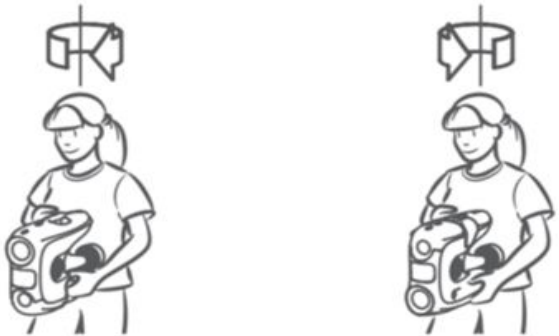
6. Then place the drone on a horizontal surface and tap “Finish calibration”. Leave the drone for a while to finish the calibration.

7. If the aircraft status indicator blinks red, the calibration has failed. Change your position and restart back the process.

8. ***“Finish calibration” button will appear when all the calibration movements are completed***

CALIBRATION COMPLETED!

Rotating horizontally



Rotating vertically



Rotating sideways



Horizontal surface



B- CALIBRATION OF ACCELEROMETER

Accelerometer calibration establishes a reference point for the drone's level position and ensures that the drone maintains stability during flight. Incorrect accelerometer calibration can lead to imprecise flight control responses, making it challenging to maintain a steady hover or execute precise maneuvers

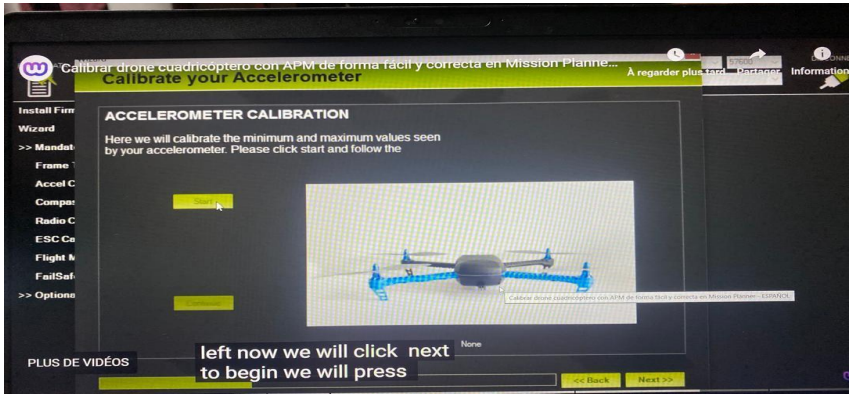
NOTE: Accelerometers must be calibrated yearly to ensure the accuracy of the data they are collecting. Calibrations are generally acceptable for a one-year term then they must be calibrated again. This process must be continued for the entire life of the sensor.

These are the steps to follow for the calibration of the accelerometer, choose an open area to carry out the following procedure;

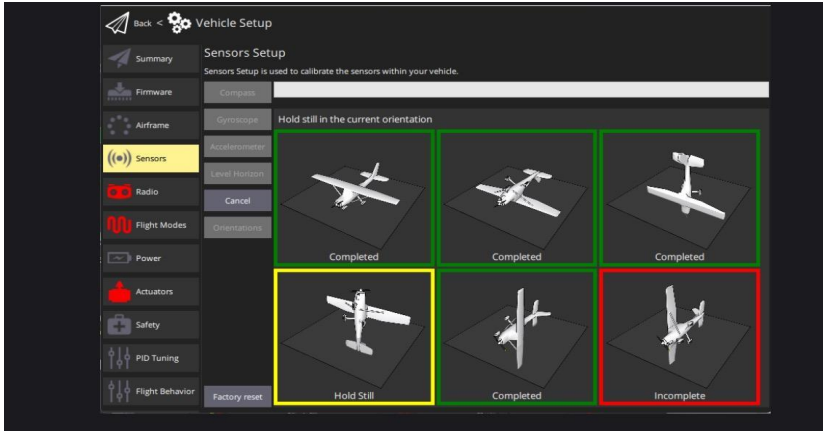
1-Step one and step two are repeated as above for the compass calibration.

2. Position the accelerometer with the arrow pointing down for the first calibration point.

3-Next we are going to select accelerometer calibration sensor button and select START then OK, then we have to follow the instructions required.



4-When clicked on start then, Position the vehicle as guided by the images on the screen. Once prompted (the orientation-image turns yellow) hold the vehicle still. Once the calibration is complete for the current orientation the associated image on the screen will turn green. Then NEXT and follow the same procedure for different positions asked.



CALIBRATION SUCCESSFUL!

c-RADIOLINK R9DS CALIBRATION AND CONNECTION

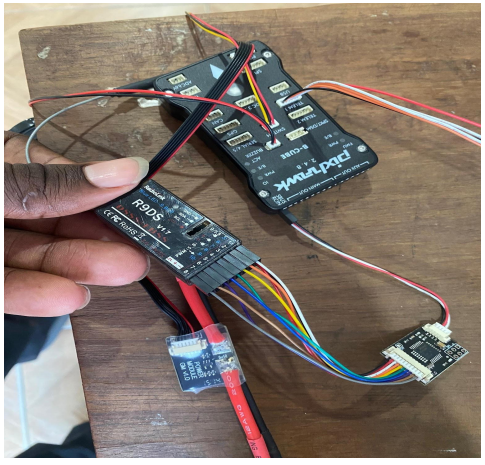
R9DS is a receiver with different channels for transmission of signals. This calibration is to connect the R9DS (receiver) to the Radio link command control (transmitter).

Pairing the signal with the transmitter:

1. Place the transmitter and receiver within one meter

2. Turn on the transmitter and then the receiver

3. The receiver has a black binding button. Press it twice within two seconds. The LED will flash. After about eight flashes, the diode will start to burn with a constant light, thus signaling the correct pairing of the transmitter with the receiver.



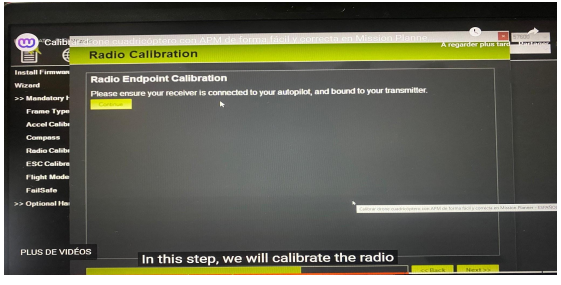




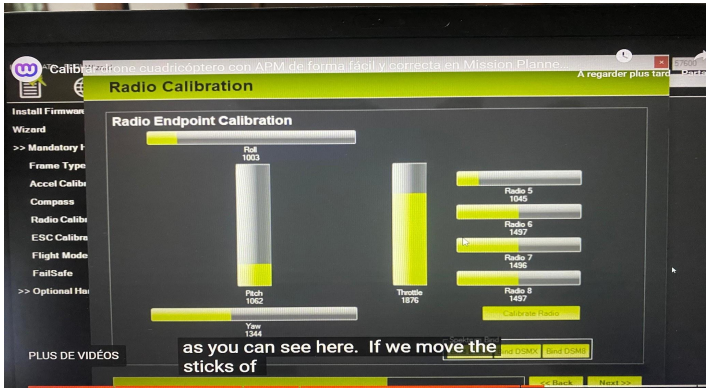
Now we’re moving to the procedure of the calibration of the radio, these are the steps to follow;

1.Click on radio calibration on the screen in the setup

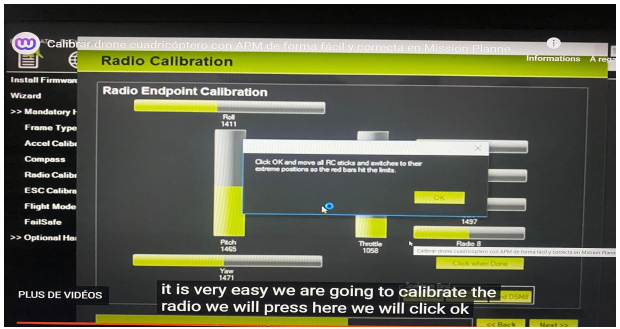
2.In this step we will calibrate the radio in our box, then pick up the station, we will turn it on and then enter and click on continue as you can see below

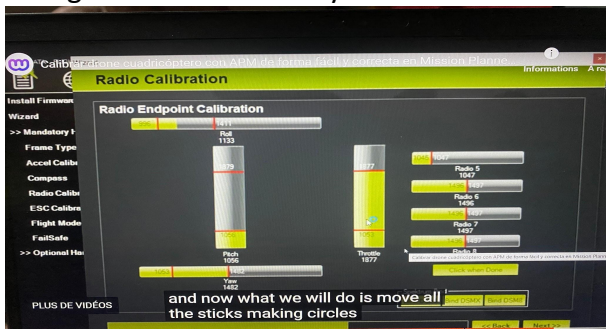


3.Now if we move the sticks of the station, the sticks broadcast planner's bars move up, down and to the sides. Before calibrating the radio, it is very important that you do not pull the right stick up and here in broadcast planner the pitch bar, when you pull the right stick up it should go down and when you pull the right stick down it should go up the pitch, if you have it like that then the radio is configured correctly. If it goes the other way round then you have to change it in the station itself.

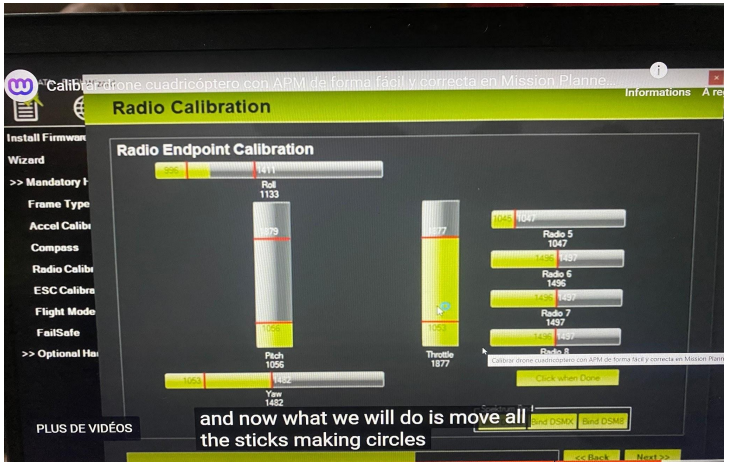


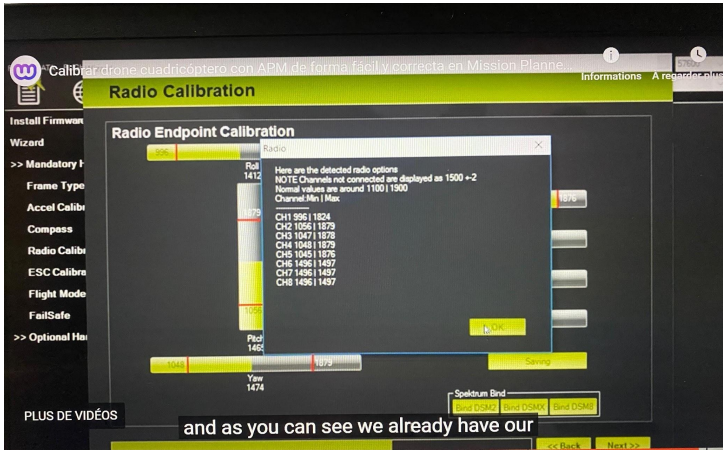
4-But then as we calibrate the radio, it is very easy. We are going to calibrate the radio. We will press on "calibrate radio" and then OK and now what we will do is to move all the sticks making circles. So that they detect the maximum positions of the station.





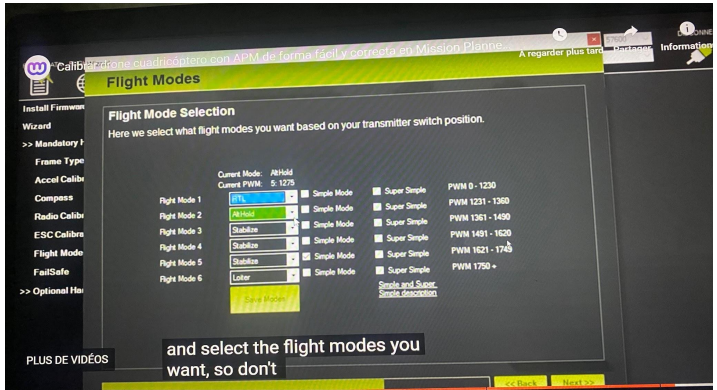
5- We will do the stick's path toward all its directions and once we have it, we will also move the flight mode selection switches so that they are also calibrated. What we have to do before clicking where we will have some sticks, the gas and direction sticks to the center so that the radio will be calibrated completely. Then going to click on “click when done” and then OK.





6-Ater the click on OK will appear the different flight mode where you have to change the flight modes on the station and select the one you want.

CALIBRATION DONE!!



D) ESCs CALIBRATION

ESC calibration will vary based on what brand of ESC you are using, so always refer to the documentation for the brand of ESC you are using for specific information (such as tones). "**All at once**" calibration works well for most ESCs, so it is a good idea to attempt it first and if that fails try the "**Manual ESC-by-ESC**" method.

NOTE: Before calibrating ESCs, please ensure that your copter has NO PROPS on it and that the autopilot is NOT CONNECTED to your computer via USB and the Lipo battery is disconnected.





Here we are going to talk about three types of ESCs calibration which are;

All at once calibration, Manual ESCs by ESCs calibration, and Semi-automatic ESCs by ESCs calibration. But the calibration that fits us was **Manual ESCs by ESCs calibration**.

Let start with the first calibration **"ALL AT ONCE CALIBRATION"**,

A) **ALL AT ONCE CALIBRATION**

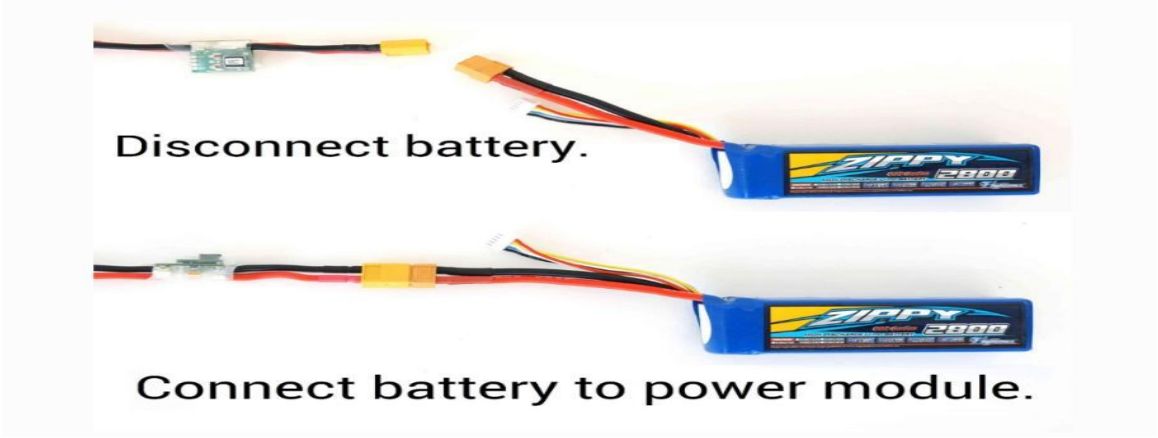
1- Turn on your transmitter and put the throttle stick at maximum.



2- Connect the Lipo battery. The autopilot red, blue and yellow LEDs will light up in a cyclical pattern. This means the it's ready to go into ESC calibration mode the next time you plug it in.



3- With the transmitter throttle stick still high, disconnect and reconnect the battery.



4- For Autopilots with a safety switch, push it until the LED displays solid red

5- The autopilot is now in ESC calibration mode.

6- Wait for your ESCs to emit the musical tone, the regular number of beeps indicating your battery's cell count (i.e. 3 for 3S, 4 for 4S) and then an additional two beeps to indicate that the maximum throttle has been captured.

7- Pull the transmitter throttle stick down to its minimum position.



8- The ESCs should then emit a long tone indicating that the minimum throttle has been captured and the calibration is complete.

9- If the long tone indicating successful calibration was heard, the ESCs are “live” now and if you raise the throttle a bit they should spin. Test that the motors spin by raising the throttle a bit and then lowering it again.

10- Set the throttle to minimum and disconnect the battery to exit ESC-calibration mode.

B) Manual ESC-by-ESC CALIBRATION

1- Plug one of your ESC three-wire cables into the throttle channel of the RC receiver. (This is usually channel 3.)

2- Turn on the transmitter and set throttle stick to maximum (full up).

3- Connect the LiPo battery

4- You will hear a musical tone then two beeps.

5- After the two beeps, lower the throttle stick to full down.

6- You will then hear a number of beeps (one for each battery cell you’re using) and finally a single long beep indicating the end points have been set and the ESC is calibrated.

7- Disconnect battery. Repeat these steps for all ESCs.

8-If it appears that the ESC’s did not calibrate then the throttle channel on the transmitter might need to be reversed.

9-If you are still having trouble after trying these methods (for example, ESCs still beep continuously) try lowering your throttle trim 50%.

C- Semi-automatic ESCs by ESCs calibration

1-Connect to the autopilot from a ground station such as the Mission Planner and set the ESC_CALIBRATION parameter to 3

2- Disconnect the battery and USB cable so the autopilot powers down

3- Connect the battery.

4- The arming tone will be played (if the vehicle has a buzzer attached)

5- If using an autopilot with a safety button (like the Pixhawk) press it until it displays solid red

6- You will hear a musical tone then two beeps.

7- A few seconds later you should hear a number of beeps (one for each battery cell you're using) and finally a single long beep indicating the end points have been set and the ESC is calibrated

8- Disconnect the battery and power up again normally and test as described below

TESTING

Once you have calibrated your ESCs, you can test them by plugging in your LiPo. Remember: no propellers!

-Ensure your transmitter's flight mode switch is set to "Stabilize Mode".

-Arm your copter

- Give a small amount of throttle. All motors should spin at about same speed and they should start at the same time. If the motors do not all start at the same time and spin at the same speed, the ESCs are still not properly calibrated.

-Disarm the drone

CALCULATING THE AUTONOMIE OF THE DRONE IN FLIGHT

This calculation allowed us to determine how long the battery will last when the drone is in flight and the energy needed for the flight. The main source of energy used to power the drone is the **battery.**

NOTE: The battery should be charged completely and should not be discharged completely before recharging. The lifespan of a battery is approximately 300 charges.

CALCULATIONS WITH BATTERY

Charge of battery(Q)= 6000mAh

Voltage (Tension (V)) =11.5v

BEC (Battery Eliminator Circuit) of an ESC=3A

Calculating the power consumption to lift

For one motor

P=IV, P=11.5(3)

=34.5W

For all motors

P=34.5*4

=138W

AUTONOMIE= Charge(Q)/I= 6/6 = 1h

So normally our battery allows a current of 6000mA for one hour

CALCULATIONS WITH GENERATORS

We connect 3 generators in series and input the voltage at 11.1V. Then connect the drone and make it flight to register the intensity of current during the flight for the calculation.





So, during the flight, V=12.5v, I=6.36+10.2+5.55

=22.1A

AUTONOMY of drone= Q/I

=6Ah/22.1A

=0.27h, Converting in minutes

= 0.27*60= 16 min

But calculating with the new battery that we made,

Q=4400mAh, I=22.1 New AUTONOMY= Q/I

=4.4Ah/22.1A

=0.199h

=0.199*60

=12 min

04

Étude dynamique du drone et choix des matériaux adaptés

domaine EEM

DOMAINE EEM(énergie et matériaux)

ce domaine a pour objectif d'expliciter les principes de vol du drone, les forces les régissant ainsi que les phénomènes physiques mis en jeu. La partie matériau aura pour objectif la prise en compte de plusieurs facteurs afin de proposer les matériaux pour les bras et les pieds du drone.

Forces, origines et principes physiques mis en jeu

toutes les forces au niveau du drone ont des origines différentes.

- **le poids** est due à la masse du drone elle même sans oublier les effets de la pesanteur forçant tous les corps soumis à son action à adhérer au sol.
- **la portance** est engendrée par le mouvement des hélices elles mêmes entraînées par les moteurs. Cela est possible à la forme des hélices permettant l'application du théorème de BERNOULLI sur les rapports pression/vitesse.
- **la poussée** est la force générée toujours par les hélices mais sous certaines conditions, notamment l'inclinaison de l'axe des moteurs
- **la traînée** est la force générée par l'air pour s'opposer au moment du drone, explicable par la 3e loi de Newton, le principe d'action-réaction

phases de vol et comportement des forces

En fonction des différentes phases de vol, les forces ainsi que leurs intensités peuvent varier ou même ne pas exister.

- **Au décollage**

Deux forces sont à recenser lors du décollage du drone, il s'agit du poids et de la portance étant donné qu'elles sont sur le même axe. Ces forces sont opposées et c'est seulement lorsque l'intensité de la portance devient supérieur à celle du poids.

- **en vol stationnaire**

en vol stationnaire, le drone est stable et en équilibre dans les airs, seulement permis par l'égalité entre le poids et la portance; cela s'explique par le principe d'inertie énoncé par Newton stipulant que la somme des forces extérieures sur un même axe sur un système en équilibre est nulle.

phases de vol et comportement des forces

- **en déplacements latéraux**

lorsqu'on est en déplacement latéral sur l’axe x, deux nouvelles forces entrent en jeu, il s’agit de: la poussée et la traction. Il existe également la possibilité d’effectuer un mouvement de rotation.

- **atterrissage**

les forces lors de l'atterrissage sont les mêmes que lors de l’ascension, sauf que la portance dans ce cas ci diminue jusqu’à s’annuler. On appelle cela MRUV décéléré.

❖ PROPRIETES DES MATERIAUX UTILISES

Matériaux	traction	Élasticité (module de Young)	Dureté	Masse volumique Kg /m3	Légèreté	rigidité	Corrosion	Thermi-Que W/mK
Fibre de carbone	2500-7000 MPa	189-240 GPa	66,9-74,1 HV	1,8	1,6 g/cm3	200-300 GPa	Corrosion faible	100
Plastique(PE)	700 MPa	0,1-0,2 GPa	60-200HV	950	1,7 g/cm3	2,3-2,5 GPa	Non Corrosif	0,2-0,22
ABS	6,1-17 MPa	2,9 GPa	60 HV	1040	1,05 g/cm3	2,3 GPa	Corrosif	0,2

Matrice multicritères de nos matériaux

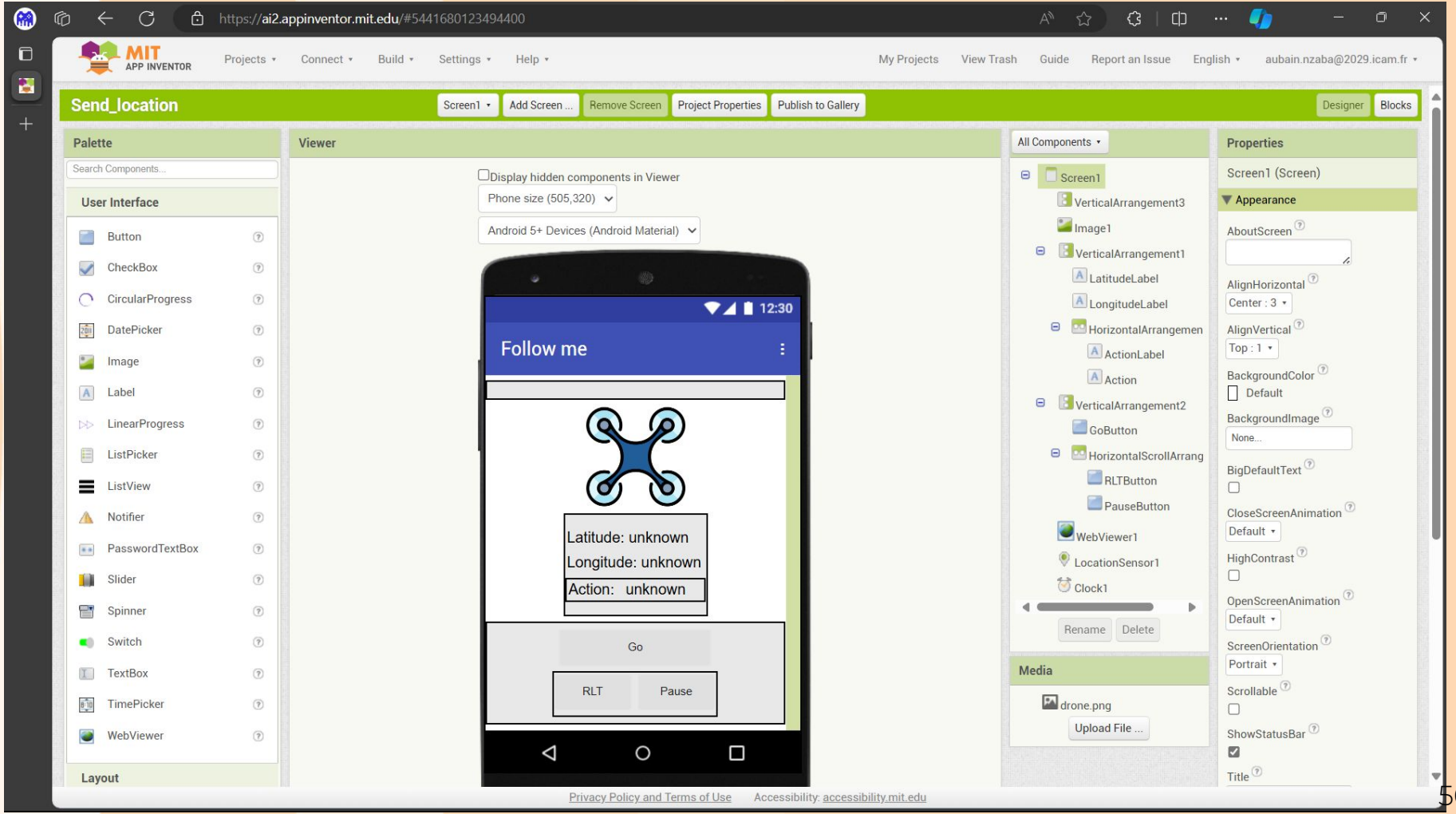
Dans notre travail, cette matrice va nous permettre de faire un choix précis et concis de nos matériaux. nous allons donc évaluer les propriétés avec une pondération allant de 1-5

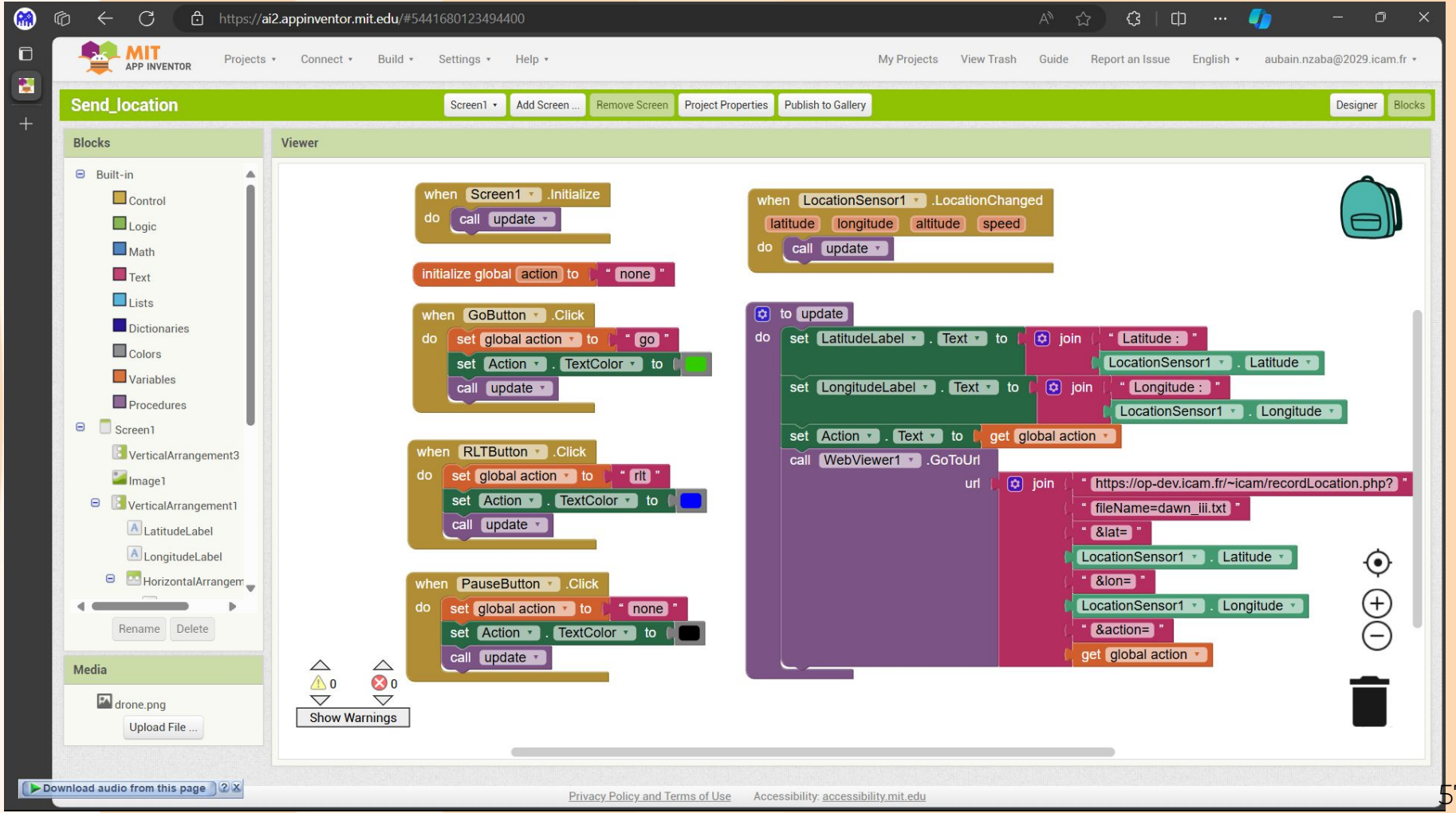
matériaux	traction	élasticité	dureté	légèreté	rigidité	Masse volumique	prix	TOTAL
Fibre de carbone	3	3	2	2	3	1	3	17
ABS	1	1	1	1	1	2	1	8
polyamide	2	3	2	2	1	2	2	14
PE	2	1	2	2	1	2	1	11

05

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DatePicker
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LinearProgress
ListPicker
ListView
Notifier
PasswordTextBox
Slider
Spinner
Switch
TextBox
TimePicker
WebViewer
Layout
Viewer
Display hidden components in Viewer
Phone size (505,320)
Android 5+ Devices (Android Material)
12:30
Follow me
Latitude: unknown
Longitude: unknown
Action: unknown
Go
RLT
Pause
All Components
Screen1
VerticalArrangement3
Image1
VerticalArrangement1
LatitudeLabel
LongitudeLabel
HorizontalArrangemen
ActionLabel
Action
VerticalArrangement2
GoButton
HorizontalScrollArrang
RLTButton
PauseButton
WebViewer1
LocationSensor1
Clock1
Rename
Delete
Media
drone.png
Upload File ...
Properties
Screen1 (Screen)
Appearance
AboutScreen
AlignHorizontal
Center : 3
AlignVertical
Top : 1
BackgroundColor
Default
BackgroundImage
None...
BigDefaultText
CloseScreenAnimation
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Screen1
Add Screen ...
Remove Screen
Project Properties
Publish to Gallery
Designer
Blocks
Blocks
Built-in
Control
Logic
Math
Text
Lists
Dictionaries
Colors
Variables
Procedures
Screen1
VerticalArrangement3
Image1
VerticalArrangement1
LatitudeLabel
LongitudeLabel
HorizontalArrangem
Rename
Delete
Media
drone.png
Upload File ...
Viewer
when Screen1 .Initialize
do call update
initialize global action to " none "
when GoButton .Click
do set global action to " go "
set Action . TextColor to
call update
when RLTButton .Click
do set global action to " rlt "
set Action . TextColor to
call update
when PauseButton .Click
do set global action to " none "
set Action . TextColor to
call update
0
0
Show Warnings
when LocationSensor1 .LocationChanged
latitude longitude altitude speed
do call update
to update
do set LatitudeLabel . Text to join " Latitude : "
LocationSensor1 . Latitude
set LongitudeLabel . Text to join " Longitude : "
LocationSensor1 . Longitude
set Action . Text to get global action
call WebViewer1 .GoToUrl
url join " https://op-dev.icam.fr/~icam/recordLocation.php? "
" fileName=dawn_iii.txt "
" &lat= "
LocationSensor1 . Latitude
" &lon= "
LocationSensor1 . Longitude
" &action= "
get global action
Download audio from this page
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Demande des coordonnées

```
def request():

    # Ask information from op-dev server
    import ssl
    context = ssl._create_unverified_context()
    import urllib.request

    try:

        url = "https://op-dev.icam.fr/~icam/dawn_iii.txt"
        response = urllib.request.urlopen(url, context=context)
        data = response.read()
        global array, last_array, change
        array = data.decode('utf-8').split()

        if array != last_array :
            print(array)

            # Store the latitude and longitude in the variables
            if array[2] == "go" and (last_array[0] != array[0] or last_array[1] != array[1]) :
                latitudes.append(float(array[0]))
                longitudes.append(float(array[1]))
                change = True

            else:
                change = False

            if last_array[2] != array[2] :
                actions.append(str(array[2]))

        else:
            change = False

        last_array = array

    except urllib.error.URLError as e:

        print("URL Error:", e.reason)
        last_array = [0,0,""]
        return
```

Connexion et armement-décollage

```python
# Set up option parsing to get connection string
import argparse
parser = argparse.ArgumentParser(description='Commands vehicle using vehicle.simple_goto.')
parser.add_argument('--connect',
                    help="Vehicle connection target string. If not specified, SITL automatically started and used.")
args = parser.parse_args()

connection_string = args.connect
sitl = None


# Start SITL if no connection string specified
if not connection_string:
    import dronekit_sitl
    request()
    sitl = dronekit_sitl.start_default(float(array[0]), float(array[1]))
    connection_string = sitl.connection_string()


# Connect to the Vehicle
print('Connecting to vehicle on: %s' % connection_string)
vehicle = connect(connection_string, wait_ready=True)


arm_and_takeoff(10)

print("Set default/target airspeed to 3")
vehicle.airspeed = 3
```

Détermine la distance entre le drone et les coordonnées cible

```python
t = 0

def check_distance():
    from geopy.distance import geodesic
    global distance, t
    distance = geodesic((vehicle.location.global_frame.lat, vehicle.location.global_frame.lon), (latitudes[t], longitudes[t])).meters
    return distance
```

Boucle principale

```
# Follow the user
while True:
    request()

    if actions[len(actions)-1] == "go" and change:

        print("Going towards target (groundspeed set to 1 m/s) ...")
        target = LocationGlobalRelative(latitudes[t], longitudes[t], 5)
        vehicle.simple_goto(target, groundspeed=1)

        #Set a waiting time as long as the drone have not joined the wanted coordinates
        while actions[len(actions) - 1] == "go" and check_distance() > 0.2 :

            request()
            print(f'Around {distance} meters left to Target, Altitude: {vehicle.location.global_relative_frame.alt}')
            time.sleep(1)

        print("Target joined")

        if t < len(latitudes):
            t = t + 1

    elif actions[len(actions) - 1] == "rlt":
        print("Returning to Launch")
        vehicle.mode = VehicleMode("RTL")

        while vehicle.location.global_relative_frame.alt > 0:
            print(" Altitude: ", vehicle.location.global_relative_frame.alt)
            time.sleep(1)

        # Close vehicle object before exiting script
        print("Close vehicle object")
        vehicle.close()

        # Shut down simulator if it was started.
        if sitl:
            sitl.stop()

        break
```


10:17 AM
Latitude : -4.79081
Longitude : 11.8571
Action: none
Go
RLT
Pause

Conclusion